PM 566 Lab 7

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Requried Package

```
library(httr)
library(xml2)
library(stringr)
library(rvest)
```

Question 1.

```
# Downloading the website
website <- xml2::read_html("https://pubmed.ncbi.nlm.nih.gov/?term=sars-cov-2")
# Finding the counts
counts <- xml2::xml_find_first(website, "/html/body/main/div[9]/div[2]/div[2]/div[1]/div[
# Turning it into text
counts <- as.character(counts)
# Extracting the data using regex (matches numbers with commas)
totalcount <- stringr::str_extract(counts, "[0-9,.]+")
# Removing commas so we can convert to numeric
totalcount <- gsub(",", "", totalcount)
totalcount <- as.numeric(totalcount)
print(totalcount)
```

```
[1] 263555
```

Bassed on the result above, there are 263555 Sars-Cov-2 papers.

Question 2.

```
# read in text, each line is a separate character
abstracts <- readLines('/Users/ldh/Library/Mobile Documents/com~apple~CloudDocs/USC PhD/S
# combine all text into one character
abstracts <- paste(abstracts, collapse = '\n')
# split the text whenever 3 new lines occur in a row (indicating two blank lines)
abstracts <- unlist(strsplit(abstracts, split = '\n\n\n'))
# replace any remaining "\n" symbols with spaces
abstracts <- gsub("\n", " ", abstracts)
# replace multiple spaces with single space
abstracts <- gsub(" +", " ", abstracts)
```

Question 3.

```
# One compact pattern for:
## "University of ...", optionally "of the ..."
## "... University"
## "... Institute of ...", optionally "of the ..."
pattern <- "\\b(?:University\\s+of\\s+(?:the\\s+)?[A-Z][\\w'&.-]+(?:\\s+[A-Z][\\w'&.-]+)*

institution <- str_extract_all(abstracts, pattern)  # extract matches
institution <- unlist(institution)                  # flatten
institution <- stringr::str_squish(institution)     # trim & compress spaces

# Frequency (top 20)
head(sort(table(institution), decreasing = TRUE), 20)
```

```
institution
               University of California                      Stanford University
                                   4375                                     1171
      University of Southern California University of California San Francisco
                                    778                                      697
     University of California San Diego                           The University
                                    673                                      410
               University of Washington                      Columbia University
                                    344                                      275
   University of California Los Angeles                 Johns Hopkins University
                                    252                                      250
                      Oxford University                        Emory University
                                    244                                      233
             University of Pennsylvania                     University of Oxford
                                    230                                      198
                  Vanderbilt University                    University of Toronto
                                    196                                      191
                       Duke University                    Washington University
                                    179                                      174
                 University of Michigan                          Yale University
                                    173                                      149
```

Based on the result above, the 10 most common institutions the reason I run 20 because there are too many universities in the UC systems.

Question 4.

```
# read in text, each line is a separate character
abstracts <- readLines('/Users/ldh/Library/Mobile Documents/com~apple~CloudDocs/USC PhD/S
# combine all text into one character
abstracts <- paste(abstracts, collapse = '\n')
# split the text whenever 3 new lines occur in a row (indicating two blank lines)
abstracts <- unlist(strsplit(abstracts, split = '\n\n\n'))
```

```
journal <- str_match(
  abstracts,
  regex("(?m)^\\s*\\d+\\.\\s*([^\\n]+)")
)[, 2]

journal <- str_squish(journal)
```

```
titles <- sapply(abstracts, function(x){
  parts <- unlist(strsplit(x, split = "\n\n+"))
  parts <- parts[nzchar(trimws(parts))]
  if (length(parts) >= 2) parts[2] else NA_character_
}, USE.NAMES = FALSE)
titles <- gsub("\\n+", " ", titles)
titles <- str_squish(titles)
```

```
authors <- sapply(abstracts, function(x){
  parts <- unlist(strsplit(x, split = "\n\n+"))
  parts <- parts[nzchar(trimws(parts))]
  if (length(parts) >= 3) parts[3] else NA_character_
}, USE.NAMES = FALSE)
authors <- gsub("\\n+", " ", authors)
authors <- str_squish(authors)
```

```
affiliations <- str_extract(
  abstracts,
  regex("\\n\\nAuthor information:.*?(?=\\n\\n|\\Z)", dotall = TRUE)
)
#
affiliations <- str_replace(affiliations, "^\\s*\\n?\\n?Author information:\\s*", "")
affiliations <- str_squish(affiliations)
```

```
papers <- data.frame(
  journal = journal,
  title   = titles,
  authors = authors,
  affiliations = affiliations,
  stringsAsFactors = FALSE
)

knitr::kable(papers[1:5, ])
```

journal	title	authors	affiliations
Cell. 2021 Sep 16;184(19):4848-4856. doi: 10.1016/j.cell.2021.08.017. Epub 2021	The origins of SARS-CoV-2: A critical review.	Holmes EC(1), Goldstein SA(2), Rasmussen AL(3), Robertson DL(4), Crits-Christoph A(5), Wertheim JO(6), Anthony SJ(7), Barclay WS(8), Boni MF(9), Doherty PC(10), Farrar J(11), Geoghegan JL(12), Jiang X(13), Leibowitz JL(14), Neil SJD(15), Skern T(16), Weiss SR(17), Worobey M(18), Andersen KG(19), Garry RF(20), Rambaut A(21).	(1)Marie Bashir Institute for Infectious Diseases and Biosecurity, School of Life and Environmental Sciences and School of Medical Sciences, The University of Sydney, Sydney, NSW 2006, Australia. Electronic address: edward.holmes@sydney.edu.au. (2)Department of Human Genetics, University of Utah, Salt Lake City, UT 84112, USA. (3)Vaccine and Infectious Disease Organization, University of Saskatchewan, Saskatoon, SK S7N 5E3, Canada. (4)MRC-University of Glasgow Centre for Virus Research, Glasgow G61 1QH, UK. (5)Department of Plant and Microbial Biology, University of California-Berkeley, Berkeley, CA 94704, USA. (6)Department of Medicine, University of California-San Diego, La Jolla, CA 92093, USA. (7)Department of Pathology, Microbiology, and Immunology, University of California-Davis School of Veterinary Medicine, Davis, CA 95616, USA. (8)Department of Infectious Disease, Imperial College, London W2 1PG, UK. (9)Center for Infectious Disease Dynamics, Department of Biology, The Pennsylvania State University, University Park, PA 16802, USA. (10)Department of Microbiology and Immunology, The University of Melbourne at the Doherty Institute, 792 Elizabeth Street, Melbourne, VIC 3000, Australia. (11)The Wellcome Trust, London NW1 2BE, UK. (12)Department of Microbiology and Immunology, University of Otago, Dunedin 9010, New Zealand; Institute of Environmental Science and Research, Wellington 5022, New Zealand. (13)Department of Biological Sciences, Xi’an Jiaotong-Liverpool University (XJTLU), Suzhou, China. (14)Department of Microbial Pathogenesis and Immunology, Texas A&M University, College Station, TX 77807, USA. (15)Department of Infectious Diseases, King’s College London, Guy’s Hospital, London SE1 9RT, UK. (16)Max Perutz Labs, Medical University of Vienna, Vienna Biocenter, Dr. Bohr-Gasse 9/3, 1030 Vienna, Austria. (17)Department of Microbiology, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA 19104, USA. (18)Department of Ecology and Evolutionary Biology, University of Arizona, Tucson, AZ 85721, USA. (19)Department of Immunology and Microbiology, The Scripps Research Institute, La Jolla, CA 92037, USA. (20)Department of Microbiology and Immunology, Tulane University School of Medicine, New Orleans, LA 70112, USA;

journal	title	authors	affiliations
			Zalgen Labs, Germantown, MD 20876, USA. (21)Institute of Evolutionary Biology, University of Edinburgh, Edinburgh EH9 3FL, UK. Electronic address: a.rambaut@ed.ac.uk.
Science. 2021 Dec 3;374(6572):abm0829. doi: 10.1126/science.abm0829. Epub 2021	mRNA vaccines induce durable immune memory to SARS-CoV-2 and variants of concern.	Goel RR(#)(1)(2), Painter MM(#)(1)(2), Apostolidis SA(#)(1)(2)(3), Mathew D(#)(1)(2), Meng W(1)(4), Rosenfeld AM(1)(4), Lundgreen KA(5), Reynaldi A(6), Khoury DS(6), Pattekar A(2), Gouma S(5), Kuri-Cervantes L(1)(5), Hicks P(5), Dysinger S(5), Hicks A(2), Sharma H(2), Herring S(2), Korte S(2), Baxter AE(1), Oldridge DA(1)(4), Giles JR(1)(7)(8), Weirick ME(5), McAllister CM(5), Awofolaju M(5), Tanenbaum N(5), Drapeau EM(5), Dougherty J(1), Long S(1), D'Andrea K(1), Hamilton JT(2)(5), McLaughlin M(1), Williams JC(2), Adamski S(2), Kuthuru O(1); UPenn COVID Processing Unit‡; Frank I(9),	(1)Institute for Immunology, University of Pennsylvania Perelman School of Medicine, Philadelphia, PA, USA. (2)Immune Health, University of Pennsylvania Perelman School of Medicine, Philadelphia, PA, USA. (3)Division of Rheumatology, University of Pennsylvania Perelman School of Medicine, Philadelphia, PA, USA. (4)Department of Pathology and Laboratory Medicine, University of Pennsylvania Perelman School of Medicine, Philadelphia, PA, USA. (5)Department of Microbiology, University of Pennsylvania Perelman School of Medicine, Philadelphia, PA, USA. (6)Kirby Institute, University of New South Wales, Sydney, NSW, Australia. (7)Department of Systems Pharmacology and Translational Therapeutics, University of Pennsylvania Perelman School of Medicine, Philadelphia, PA, USA. (8)Parker Institute for Cancer Immunotherapy, University of Pennsylvania Perelman School of Medicine, Philadelphia, PA, USA. (9)Division of Infectious Disease, University of Pennsylvania Perelman School of Medicine, Philadelphia, PA, USA. (10)Division of Infectious Disease, Department of Pediatrics, Children's Hospital of Philadelphia, Philadelphia, PA, USA. (11)Center for Infectious Disease and Vaccine Research, La Jolla Institute for Immunology (LJI), La Jolla, CA, USA. (12)Department of Medicine, Division of Infectious Diseases and Global Public Health, University of California San Diego (UCSD), La Jolla, CA, USA. (#)Contributed equally

journal	title	authors	affiliations
		Betts MR(1)(5), Vella LA(10), Grifoni A(11), Weiskopf D(11), Sette A(11)(12), Hensley SE(5), Davenport MP(6), Bates P(5), Luning Prak ET(1)(4), Greenplate AR(1)(2), Wherry EJ(1)(2)(7)(8).	
Occup Environ Med. 2021 Nov;78(11):789-792. doi: 10.1136/oemed-2021-107461. Epub	SARS-CoV-2 antibody seroprevalence among firefighters in Orange County, California.	Vieira V(1)(2), Tang IW(3), Bartell S(3)(2), Zahn M(4), Fedoruk MJ(3)(2).	(1)Department of Environmental and Occupational Health, University of California Irvine Susan and Henry Samueli College of Health Sciences, Irvine, California, USA vvieira@uci.edu. (2)University of California Irvine Center for Occupational and Environmental Health, Irvine, California, USA. (3)Department of Environmental and Occupational Health, University of California Irvine Susan and Henry Samueli College of Health Sciences, Irvine, California, USA. (4)Orange County Health Care Agency, Santa Ana, California, USA.
Science. 2021 Aug 27;373(6558):eabd9149. doi: 10.1126/science.abd9149.	Airborne transmission of respiratory viruses.	Wang CC(1)(2)(3)(4), Prather KA(5), Sznitman J(6), Jimenez JL(6)(7), Lakdawala SS(8), Tufekci Z(9), Marr LC(3)(10).	(1)Department of Chemistry, National Sun Yat-sen University, Kaohsiung, Taiwan 804, Republic of China. chiawang@mail.nsysu.edu.tw kprather@ucsd.edu. (2)Scripps Institution of Oceanography, University of California San Diego, La Jolla, CA 92037, USA. (3)Aerosol Science Research Center, National Sun Yat-sen University, Kaohsiung, Taiwan 804, Republic of China. (4)Department of Chemistry, National Sun Yat-sen University, Kaohsiung, Taiwan 804, Republic of China. (5)Scripps Institution of Oceanography, University of California San Diego, La Jolla, CA 92037, USA. chiawang@mail.nsysu.edu.tw kprather@ucsd.edu. (6)Department of Biomedical Engineering, Israel Institute of Technology, Haifa 32000, Israel. (7)Department of Chemistry and CIRES, University of Colorado, Boulder, CO 80309, USA. (8)Department of Microbiology and Molecular Genetics, University of Pittsburgh School of Medicine, Pittsburgh, PA 15219, USA. (9)School of Information and Department of Sociology, University of North Carolina, Chapel Hill, NC 27599, USA. (10)Department of Civil and Environmental

journal	title	authors	affiliations
			Engineering, Virginia Tech, Blacksburg, VA 24061, USA.
Sci Rep. 2021 Feb 4;11(1):3081. doi: 10.1038/s41598-021-82662-x.	Estimated seroprevalence of SARS-CoV-2 antibodies among adults in Orange County, California.	Bruckner TA(1), Parker DM(2), Bartell SM(2)(3), Vieira VM(2), Khan S(4), Noymer A(2), Drum E(2), Albala B(5), Zahn M(6), Boden-Albala B(7).	(1)Program in Public Health, University of California, Irvine, 653 E. Peltason Dr, Irvine, CA, 92697, USA. tim.bruckner@uci.edu. (2)Program in Public Health, University of California, Irvine, 653 E. Peltason Dr, Irvine, CA, 92697, USA. (3)Department of Statistics, University of California, Irvine, Bren Hall 2019, Irvine, CA, 92697-1250, USA. (4)School of Medicine, University of California, Irvine, 1001 Health Sciences Rd, Irvine, CA, 92697, USA. (5)Center for Clinical Research, School of Medicine, University of California, Irvine, 1001 Health Sciences Rd, Irvine, CA, 92617, USA. (6)Orange County Health Care Agency, 405 W. 5th St., Santa Ana, CA, 92701, USA. (7)Program in Public Health, University of California, Irvine, 653 E. Peltason Dr, Irvine, CA, 92697, USA. BBODENAL@hs.uci.edu.

Above is my result of this question